

1. (a) Derive an expression for V_x in Eq. (2.11) of Patankar (1980).
(b) Show that V_x becomes zero when the density and viscosity are constant. (Use the continuity equation).

(a)

In Patankar's book, he write the momentum equation as follows.

$$\frac{\partial}{\partial t}(\rho u) + \text{div}(\rho \tilde{u} u) = \text{div}(\mu \text{grad } u) - \frac{\partial p}{\partial x} + B_x + V_x \quad (x\text{-direction momentum equation})$$

where u denotes the x -direction velocity, B_x is the x -direction body force per unit volume, and

V_x stands for the viscous terms that are in addition to those expressed by $\text{div}(\mu \text{grad } u)$.

The stress in a static fluid must be of the form $T_{ij} = -p \delta_{ij}$ (δ_{ij} : Kronecker delta)

A moving fluid develops additional stress components, τ_{ij} , because of viscosity.

The stress tensor called the deviatoric stress tensor (τ_{ij}) must depend on the velocity gradient tensor $\frac{\partial u_i}{\partial x_j}$.

We know that velocity gradient tensor is composed of strain rate tensor and rotation tensor.

$$\frac{\partial u_i}{\partial x_j} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right)$$

$$\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) : \text{strain tensor } S_{ij} \quad (S_{ij} = S_{ji}) \quad , \quad \frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} : \text{rotation tensor } R_{ij} \quad (R_{ij} = -R_{ji})$$

However, by definition, stresses only develop in fluid elements that change shape.

Viscous stress deforms a fluid element and is not related with rotation tensor.

Therefore, only the symmetric part ($S_{ij} = S_{ji}$) should be considered.

Let's assume that there is a linear relationship between viscous stress and velocity gradient tensor.

$$\tau_{ij} = K_{ijmn} \cdot \frac{\partial u_m}{\partial x_n} = K_{ijmn} (S_{mn}) \quad , \quad \tau_{ij} = \tau_{ji} \quad (K_{ijmn} : \text{fourth-order tensor having 81 components})$$

And also assume isotropic fluid, the fourth order tensor can be represented as follows.

$$K_{ijmn} = \lambda \delta_{ij} \delta_{mn} + \mu \delta_{im} \delta_{jn} + \gamma \delta_{in} \delta_{jm} \quad (\lambda, \mu, \text{ and } \gamma \text{ are scalars that depend on the local thermodynamics.})$$

Viscous stress tensor is symmetric, so we can switch the indices ($i \leftrightarrow j$)

$$K_{ijmn} = K_{jinm} = \lambda \delta_{ji} \delta_{mn} + \mu \delta_{jm} \delta_{in} + \gamma \delta_{jn} \delta_{im}$$

Because of the symmetric property, we can know that μ and γ have to be equal to each other.

$$K_{ijmn} = \lambda \delta_{ij} \delta_{mn} + \mu (\delta_{im} \delta_{jn} + \delta_{in} \delta_{jm})$$

$$\begin{aligned} \tau_{ij} &= \lambda \delta_{ij} S_{mn} S_{mn} + \mu (\delta_{im} \delta_{jn} S_{mn} + \delta_{in} \delta_{jm} S_{mn}) \\ &= \lambda \delta_{ij} S_{mn} S_{mn} + \mu (S_{ij} + S_{ji}) \\ &= \lambda S_{mn} \delta_{ij} + 2\mu S_{ij} = \lambda (\nabla \cdot \tilde{u}) \delta_{ij} + 2\mu S_{ij} \end{aligned}$$

$$** \quad a_i \delta_{ij} = a_j \rightarrow a_1 \delta_{11} + a_2 \delta_{21} + a_3 \delta_{31}$$

$$\begin{aligned} j=1 &\rightarrow a_1 \delta_{11} = a_1 \\ j=2 &\rightarrow a_2 \delta_{22} = a_2 \end{aligned}$$

$$\rho \frac{D\tilde{u}}{Dt} = -\nabla p + \nabla \cdot \tilde{\tau} + f_b$$

$$\begin{aligned} \nabla \cdot \tilde{\tau} &\rightarrow \frac{\partial \tau_{ij}}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\lambda (\nabla \cdot \tilde{u}) \delta_{ij} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right) \\ &= \lambda \frac{\partial}{\partial x_j} (\nabla \cdot \tilde{u}) + \left(\mu \frac{\partial^2 u_i}{\partial x_j^2} + \frac{\partial \mu}{\partial x_j} \frac{\partial u_i}{\partial x_j} \right) + \left(\mu \frac{\partial^2 u_j}{\partial x_j \partial x_i} + \frac{\partial \mu}{\partial x_j} \cdot \frac{\partial u_j}{\partial x_i} \right) \\ &= \nabla \{ \lambda (\nabla \cdot \tilde{u}) \} + \nabla \cdot \{ \mu (\nabla \tilde{u} + (\nabla \tilde{u})^T) \} \end{aligned}$$

Thus the momentum equation becomes

$$\rho \frac{D\tilde{u}}{Dt} = -\nabla p + \nabla \cdot \{ \mu \cdot \nabla \tilde{u} \} + \nabla \cdot \{ \mu (\nabla \tilde{u})^T \} + \nabla (\lambda \nabla \cdot \tilde{u}) + f_b$$

$$x\text{-dir} : \frac{\partial}{\partial t} (\rho u) + \text{div}(\rho \tilde{u} u) = - \frac{\partial p}{\partial x} + \text{div}(\mu \text{grad} u) + \text{div}(\mu (\text{grad} u)^T) + \text{grad}(\lambda \cdot \text{div}(\tilde{u})) + B_x$$

$$\therefore V_x = \text{div}(\mu (\text{grad} u)^T) + \text{grad}(\lambda \cdot \text{div}(u)) \quad (u : x\text{-direction velocity})$$

(b)

When the density and viscosity are constant, the continuity equation and the momentum equation take the following form.

$$\text{continuity} : \frac{\partial \rho}{\partial t} + \nabla \cdot \tilde{u} = 0 \rightarrow \rho = \text{constant} \rightarrow \nabla \cdot \tilde{u} = 0$$

$$\text{momentum} : \rho \frac{D\tilde{u}}{Dt} = -\nabla p + \nabla \cdot \{ \mu \cdot \nabla \tilde{u} \} + \nabla \cdot \{ \mu (\nabla \tilde{u})^T \} + \nabla (\lambda \underbrace{\nabla \cdot \tilde{u}}_{=V_x}) + B_x$$

$$\rho, \mu : \text{constant} \rightarrow \nabla (\lambda \underbrace{\nabla \cdot \tilde{u}}_{=0}) = 0, \quad \nabla \cdot \{ \mu (\nabla \tilde{u})^T \} = \mu \nabla \cdot (\nabla \tilde{u})^T$$

Here, we can convert $\nabla \cdot (\nabla \tilde{u})^T$ to $\nabla (\nabla \cdot \tilde{u})$ by the smooth assumption. ($\partial x_j \leftrightarrow \partial x_i$)

$$\frac{\partial}{\partial x_j} \left(\frac{\partial u_i}{\partial x_i} \right) = \frac{\partial}{\partial x_i} \left(\frac{\partial u_j}{\partial x_j} \right) = \frac{\partial}{\partial x_i} (\nabla \cdot \tilde{u}) = \nabla (\nabla \cdot \tilde{u})$$

$$\nabla \cdot (\nabla \tilde{u})^T = \nabla (\underbrace{\nabla \cdot \tilde{u}}_{=0}) = 0 \rightarrow \nabla \cdot \{ \mu (\nabla \tilde{u})^T \} = 0$$

$$\Rightarrow \rho \frac{D\tilde{u}}{Dt} = -\nabla p + \nabla \cdot \{ \mu \nabla \tilde{u} \} + f_B$$

$$x\text{-dir} : \rho \frac{\partial u}{\partial t} + \rho \text{div}(\tilde{u} u) = - \frac{\partial p}{\partial x} + \mu \text{div}(\text{grad} u) + B_x$$

Thus, the additional viscous term, V_x , becomes zero when the density and viscosity are constant.

2. Derive the canonical form for elliptic equations

$$\phi_{\xi\xi} + \phi_{\eta\eta} = h(\phi_\xi, \phi_\eta, \phi, \xi, \eta)$$

by applying the transformation

$$\xi = y - c_1 x \quad \eta = c_2 x$$

to the following general second-order PDE

$$a\phi_{xx} + b\phi_{xy} + c\phi_{yy} + d\phi_x + e\phi_y + f\phi = g(x, y)$$

Note that c_1 and c_2 need to be properly determined for the desired transformation.

$$\frac{\partial \xi}{\partial x} = \frac{\partial}{\partial x} (y - c_1 x) = -c_1, \quad \frac{\partial \eta}{\partial x} = \frac{\partial}{\partial x} (c_2 x) = c_2,$$

$$\frac{\partial \xi}{\partial y} = \frac{\partial}{\partial y} (y - c_1 x) = 1, \quad \frac{\partial \eta}{\partial y} = 0$$

$$\phi_x = \frac{\partial \phi}{\partial x} = \frac{\partial \phi}{\partial \xi} \cdot \frac{\partial \xi}{\partial x} + \frac{\partial \phi}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} = \frac{\partial \phi}{\partial \xi} (-c_1) + \frac{\partial \phi}{\partial \eta} \cdot c_2$$

$$\phi_y = \frac{\partial \phi}{\partial y} = \frac{\partial \phi}{\partial \xi} \cdot \frac{\partial \xi}{\partial y} + \frac{\partial \phi}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} = \frac{\partial \phi}{\partial \xi} \cdot 1$$

$$\begin{aligned} \phi_{xx} = \frac{\partial \phi_x}{\partial x} &= \frac{\partial \phi_x}{\partial \xi} \cdot \frac{\partial \xi}{\partial x} + \frac{\partial \phi_x}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} = \frac{\partial}{\partial \xi} \left(-c_1 \cdot \frac{\partial \phi}{\partial \xi} + c_2 \cdot \frac{\partial \phi}{\partial \eta} \right) \cdot (-c_1) + \frac{\partial}{\partial \eta} \left(-c_1 \cdot \frac{\partial \phi}{\partial \xi} + c_2 \cdot \frac{\partial \phi}{\partial \eta} \right) \cdot c_2 \\ &= c_1^2 \phi_{\xi\xi} - 2c_1 c_2 \phi_{\xi\eta} + c_2^2 \phi_{\eta\eta} \end{aligned}$$

$$\phi_{xy} = \frac{\partial \phi_x}{\partial y} = \frac{\partial \phi_x}{\partial \xi} \cdot \frac{\partial \xi}{\partial y} + \frac{\partial \phi_x}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} = \frac{\partial}{\partial \xi} \left(-c_1 \cdot \frac{\partial \phi}{\partial \xi} + c_2 \cdot \frac{\partial \phi}{\partial \eta} \right) \cdot 1 + 0 = -c_1 \phi_{\xi\xi} + c_2 \phi_{\xi\eta}$$

$$\phi_{yy} = \frac{\partial \phi_y}{\partial y} = \frac{\partial \phi_y}{\partial \xi} \cdot \frac{\partial \xi}{\partial y} + \frac{\partial \phi_y}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} = \frac{\partial}{\partial \xi} \left(\frac{\partial \phi}{\partial \xi} \right) \cdot 1 + \frac{\partial}{\partial \eta} \left(\frac{\partial \phi}{\partial \xi} \right) \cdot 0 = \phi_{\xi\xi}$$

Our general second-order PDE is

$$a\phi_{xx} + b\phi_{xy} + c\phi_{yy} + d\phi_x + e\phi_y + f\phi = g(x, y)$$

$$\begin{aligned} \longrightarrow & a(c_1^2 \phi_{\xi\xi} - 2c_1 c_2 \phi_{\xi\eta} + c_2^2 \phi_{\eta\eta}) + b(-c_1 \phi_{\xi\xi} + c_2 \phi_{\xi\eta}) + c(\phi_{\xi\xi}) \\ & + d \left\{ \frac{\partial \phi}{\partial \xi} (-c_1) + \frac{\partial \phi}{\partial \eta} \cdot c_2 \right\} + e \left\{ \frac{\partial \phi}{\partial \xi} \right\} + f\phi = g(x, y) \end{aligned}$$

$$g(x, y) = (ac_1^2 - bc_1 + c) \phi_{\xi\xi} + (-2ac_1 c_2 + bc_2) \phi_{\xi\eta} + ac_2^2 \phi_{\eta\eta} + (-dc_1 + e) \phi_\xi + c_2 \phi_\eta + f\phi$$

For a canonical elliptic form, the coefficient of $\phi_{\xi\eta}$ has to be zero.

$$-2ac_1 c_2 + bc_2 = 0 \longrightarrow c_1 = \frac{b}{2a}$$

The coefficient of $\phi_{\xi\xi}$ becomes

$$ac_1^2 - bc_1 + c = a \cdot \frac{b^2}{4a^2} - b \cdot \frac{b}{2a} + c = -\frac{b^2}{4a} + c$$

We want that the coefficient of $\phi_{\eta\eta}$ is equal to $\phi_{\xi\xi}$

$$-\frac{b^2}{4a} + c = ac_2^2 \longrightarrow c_2^2 = \frac{4ac - b^2}{4a^3}, \quad c_2 = \frac{\sqrt{4ac - b^2}}{2a^{3/2}} \quad (\because \text{elliptic: } b^2 - 4ac < 0, a > 0, c > 0)$$

Therefore,

$$a_2 (\phi_{\xi\xi} + \phi_{\eta\eta}) = g(x,y) - (-dc_1 + e)\phi_{\xi} - c_2\phi_{\eta} - f\phi$$

$$\therefore \phi_{\xi\xi} + \phi_{\eta\eta} = \frac{1}{a_2} \left\{ g(x,y) + (dc_1 - e)\phi_{\xi} - c_2\phi_{\eta} - f\phi \right\}$$

x and y must be expressed in terms of ξ and η . ($\because x = \frac{1}{c_2}\eta$, $y = \xi + \frac{c_1}{c_2}\eta$)

So, we can say that the right hand term is the function of $\xi, \eta, \phi, \phi_{\xi}$, and ϕ_{η} .

Thus, the canonical form for a second-order elliptic equation is as follows.

$$\therefore \phi_{\xi\xi} + \phi_{\eta\eta} = h(\xi, \eta, \phi, \phi_{\xi}, \phi_{\eta})$$

2. Classify the following PDEs:

$$(a) \frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} = -e^{-kt}$$

General form of Second order PDE

$$(b) \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 4$$

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

$$(c) \begin{cases} \beta^2 \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} = 0 \\ \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \end{cases}$$

$$(a) \frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} = -e^{-kt} \rightarrow \text{second order inhomogeneous PDE}$$

We need to check the sign of the discriminant.

$$b^2 - 4ac = 0 - 4 \cdot 1 \cdot 1 < 0 : \text{elliptic PDE}$$

$$(b) \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 4 \rightarrow \text{second order inhomogeneous PDE}$$

We need to check the sign of the discriminant.

$$b^2 - 4ac = 1 - 4 \cdot 1 \cdot 1 < 0 : \text{elliptic PDE}$$

$$(c) \begin{cases} \beta^2 \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} = 0 \quad \dots (1) \\ \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 \quad \dots (2) \end{cases} \xrightarrow{\frac{\partial (1)}{\partial x}, \frac{\partial (2)}{\partial y}} \begin{cases} \beta^2 \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 v}{\partial x \partial y} = 0 \quad \dots (3) \\ \frac{\partial^2 v}{\partial y \partial x} - \frac{\partial^2 u}{\partial y^2} = 0 \quad \dots (4) \end{cases}$$

$$(3) + (4) \rightarrow \beta^2 \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$$

$$D = b^2 - 4ac = 0 - 4 \cdot \beta^2 \cdot (-1) > 0 : \text{hyperbolic PDE}$$